

Implementation of Ultrasonic Energy Mapping on FPGA for Damage Detection of Composite Structure

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Abstract

Ultrasonic energy mapping (UEM) is used for quantitative evaluation of damages in various structures. This computationally intensive algorithm makes use of multi-dimensional Fourier transform and wavenumber domain filtering to capture the energy scattering waves. Field programmable gate arrays (FPGA) based parallel architecture is used to get accelerated results for damage detection. In FPGA, multi-dimensional Fourier transforms are challenging to implement since memory access is necessary to handle large data set that cannot fit on single FPGA chip. Our design is based on row-column decomposition, where 1D Fourier transform is made along rows and subsequently along columns. Two DDR3 SDRAM memories were utilized for memory access and Xilinx intellectual property cores were used for 1D Fourier transform. In this paper, we implement UEM algorithm and propose multi-dimensional Fourier transform architecture in FPGA using two DDR3 memories. The algorithm was integrated with FPGA-based Ultrasonic Propagation Imager (UPI) system. The design was synthesized on Virtex 6 XC6VLX240T chip and tested on composite structure to visualize the impact damage.

Keywords: Non-destructive evaluation, Ultrasonic Propagation Imager, Ultrasonic energy mapping, Field programmable gate array.

1. INTRODUCTION

The ultrasonic energy mapping (UEM) is a powerful method used in ultrasonic propagation imaging (UPI) system to obtain a static map for damage detection. This technique is very useful for measurable assessment of damages in the structure. UEM method involve two dimensional Fourier transform and wavenumber filtering of a full Wavefield data [1]. Previously, this algorithm was not integrated in UPI system, therefore, it used to take extra time and effort to convert the scan data and perform post processing in Matlab. To use UEM method in real field applications, we need to locally process the algorithm in the UPI system. Currently, we use Field programmable gate arrays (FPGA) based digitizer for dedicated data acquisition and signal processing.

Therefore, to integrate UEM method in UPI system, we must implement UEM technique in FPGA first. Implementation of UEM in FPGA is challenging because multi-dimensional Fourier transform for a large number of data set cannot fit into a single chip and therefore, memory access is required [2]. In this study, we propose a design to implement UEM technique on FPGA platforms for novel UPI system.

2. ULTRASONIC ENERGY MAPPING IN FPGA

The novel UPI system acquires full Wavefield data using laser scanning system and ultrasonic sensing system [3]. This ultrasonic wave field data is processed for UEM using two DDR3 SDRAM memories, where it is read and written multiple times from one memory to another, based on row-column decomposition method. Xilinx IP cores for discrete Fourier transform (DFT) and fast Fourier transform (FFT) are used to perform 1d Fourier transformation. Each IP core has different limitation for its available transformation lengths, therefore we have used both IP cores to achieve maximum flexibility for inspection area in our design. Figure 1 shows the complete flow chart for UEM implementation.

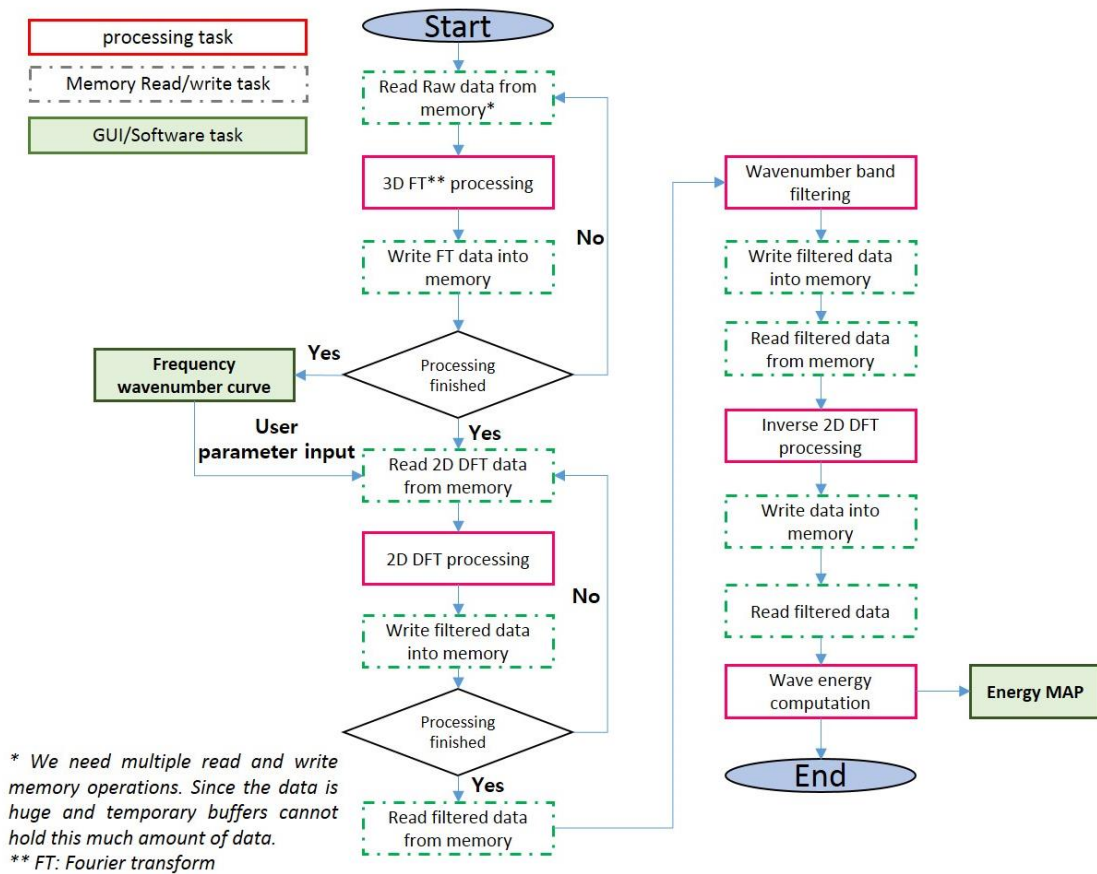


Fig. 1. Flow chart for UEM implementation in FPGA.

The processes in the flow chart are divided into processing tasks, memory read/write tasks, and GUI software tasks. Processing tasks are operations performed on the data received from the memory. Memory read/write tasks are used to fetch the data from the memory into the temporary buffer. GUI/Software tasks are used to obtain user input or display information. First of all, Wavenumber vs frequency curves are generated using 3D Fourier transform. These curves are used to determine the central wavenumber for filtering. After taking user inputs, UEM is performed and wave energy is calculated at each point to obtain an intensity map.

3. RESULTS

We target Xilinx Virtix-6 XC6VLX240T device to implement our design. To validate the UEM operation on FPGA, an experiment is performed on composite structure. The Energy map is generated for time range between 60 us to 120 us, whereas the central wavenumber applied to scan area is 0.13 mm^{-1} . Figure 2 shows the experiment setup and UEM results comparison between FPGA and Matlab. The impact damage is visible in Fig. 2 (b) where a strong wave energy field is trapped in $20 \times 20 \text{ mm}^2$ area, showing the location and size of the damage.

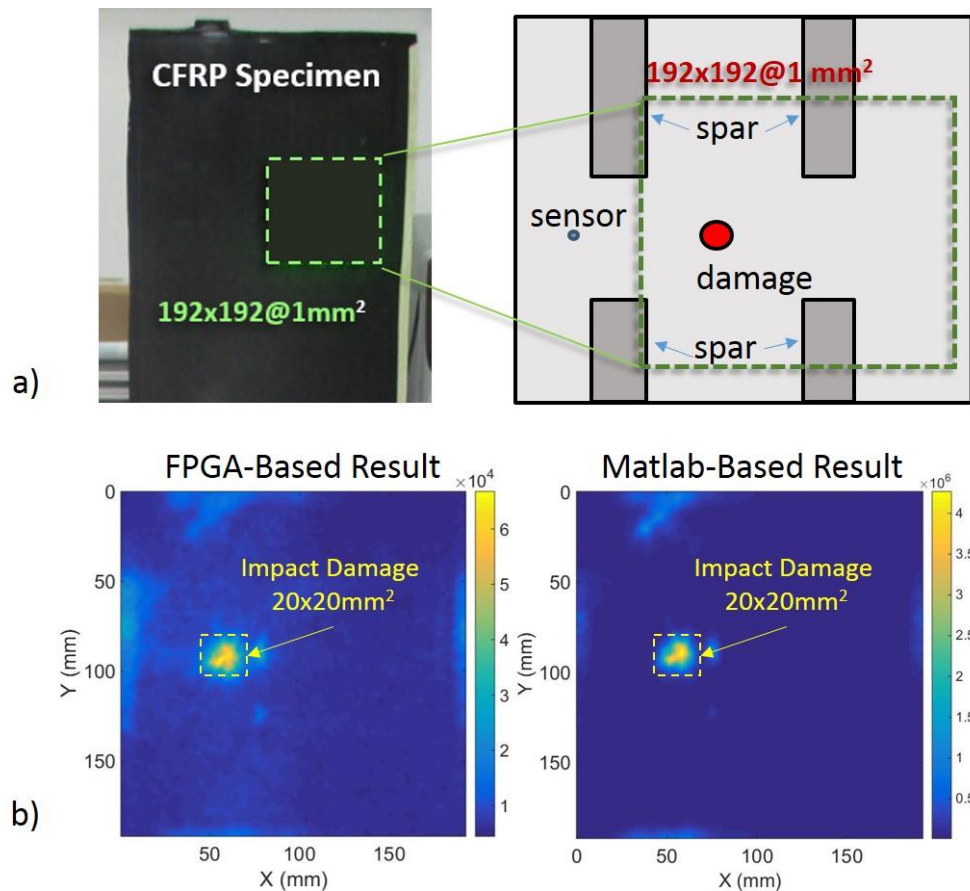


Fig. 2. (a) Experiment setup with CFRP wing box specimen, (b) UEM results generated using FPGA and Matlab

A performance graph is shown in Fig. 3 which presents the processing time taken to generate damage visualization results based on UEM method using FPGA. The result shows that it takes less than 4 seconds to process 200 x 200 mm² area having 1 mm scan point interval and total 512 samples per signal.

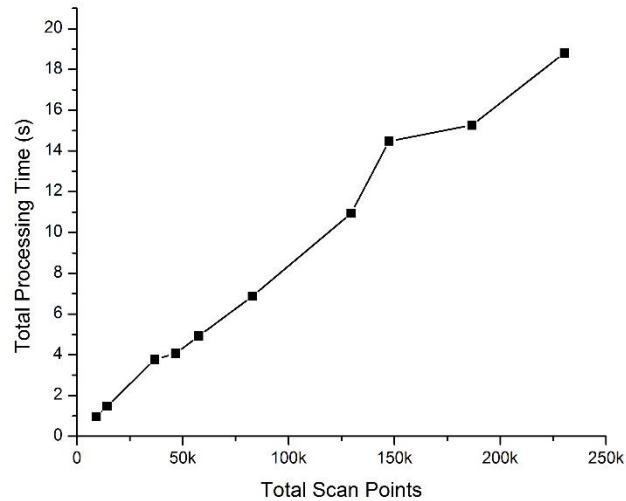


Fig. 3. Performance graph for UEM in FPGA

4. CONCLUSION

This paper showed the implementation of UEM method in FPGA. The experiment performed on a composite wing box specimen confirmed that UEM result processed in FPGA was similar with Matlab based UEM results and was able to detect 20 x 20 mm² damage. Results showed that it takes less than 4 seconds to process around 40,000 scan points. These results have proved that UEM method is successfully implemented in FPGA and integrated with UPI system for real-world applications.

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REFERENCES

- [1] Lee J. R., Cho C. M., Park C. Y., Truong C. T., Shin H. J., Jeong H., and Flynn E. B., "Spar disband visualization in in-service composite UAV with ultrasonic propagation imager", *Aerospace Science and Technology*, Vol. 45, 2015: pp. 180–185.
- [2] Yu C., Irick K., and Narayanan C., "Multidimensional DFT IP generator for FPGA platforms", *IEEE Transactions on Circuits and Systems*, Vol. 58, No. 4 April 2011: pp.755-764.
- [3] Lee J. R., Chia, C. C., Shin H. J., Park C. Y., and Yoon D. J., "Laser ultrasonic propagation imaging method in the frequency domain based on wavelet transformation", *Optics and Laser in Engineering*, Vol. 49, 2011, pp.167-175.